

$l_q^{(1)} i_{12} i_{13} i_{14} \rightarrow$

$$\begin{aligned} & \frac{1}{16\pi^2} \left(-\frac{1}{144} \frac{1}{m_\Phi^2} \left(g_2^2 c_Y^2 \overline{y}_u^{i4p} y_u^{i3p} \delta_{i1i2} + s_Y^2 \overline{y}_d^{i4p} y_d^{i3p} \left(-18 c_Y^2 \overline{y}_e^{i2r} y_e^{i1r} + g_2^2 \delta_{i1i2} \right) + \right. \right. \\ & \quad \left. s_Y^2 \overline{y}_e^{i2p} y_e^{i1p} \left(18 c_Y^2 \overline{y}_u^{i4r} y_u^{i3r} + g_2^2 \delta_{i3i4} \right) \right) + \\ & \frac{1}{9} g_2^4 \text{LF}_{3,0}[m_2] \delta_{i1i2} \delta_{i3i4} + \frac{1}{6} g_2^4 \text{LF}_{4,-1}[m_2] \delta_{i1i2} \delta_{i3i4} - \frac{8}{45} g_2^4 \text{LF}_{5,-2}[m_2] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{18} \sum_p g_2^4 \text{LF}_{3,0}[m_l^p] \delta_{i1i2} \delta_{i3i4} - \frac{5}{48} \sum_p g_2^4 \text{LF}_{4,-1}[m_l^p] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{2}{45} \sum_p g_2^4 \text{LF}_{5,-2}[m_l^p] \delta_{i1i2} \delta_{i3i4} + \frac{1}{6} \sum_p g_2^4 \text{LF}_{3,0}[m_q^p] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{5}{16} \sum_p g_2^4 \text{LF}_{4,-1}[m_q^p] \delta_{i1i2} \delta_{i3i4} + \frac{2}{15} \sum_p g_2^4 \text{LF}_{5,-2}[m_q^p] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{4} s_Y^2 c_Y^2 \left(\overline{y}_d^{i4p} y_d^{i3p} \overline{y}_e^{i2r} y_e^{i1r} - \overline{y}_e^{i2p} y_e^{i1p} \overline{y}_u^{i4r} y_u^{i3r} \right) \text{LF}_{1,2}[m_\Phi] + \\ & \frac{1}{24} \left(-g_2^2 c_Y^2 \overline{y}_u^{i4p} y_u^{i3p} \delta_{i1i2} + s_Y^2 \overline{y}_d^{i4p} y_d^{i3p} \left(3 s_Y^2 \overline{y}_e^{i2r} y_e^{i1r} - g_2^2 \delta_{i1i2} \right) + \right. \\ & \quad \left. s_Y^2 \overline{y}_e^{i2p} y_e^{i1p} \left(3 c_Y^2 \overline{y}_u^{i4r} y_u^{i3r} - g_2^2 \delta_{i3i4} \right) \right) \text{LF}_{2,1}[m_\Phi] + \\ & \frac{1}{72} g_2^2 \left(3 \left(s_Y^2 \overline{y}_d^{i4p} y_d^{i3p} + c_Y^2 \overline{y}_u^{i4p} y_u^{i3p} \right) \delta_{i1i2} + \left(3 s_Y^2 \overline{y}_e^{i2p} y_e^{i1p} + 4 g_2^2 \delta_{i1i2} \right) \delta_{i3i4} \right) \\ & \quad \text{LF}_{3,0}[m_\Phi] - \frac{5}{48} g_2^4 \text{LF}_{4,-1}[m_\Phi] \delta_{i1i2} \delta_{i3i4} + \frac{2}{45} g_2^4 \text{LF}_{5,-2}[m_\Phi] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{18} g_2^4 \text{LF}_{3,0}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{12} g_2^4 \text{LF}_{4,-1}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} - \frac{4}{45} g_2^4 \text{LF}_{5,-2}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{96} g_1^2 g_2^2 \text{LF}_{2,1,0}[m_1, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{96} g_1^2 g_2^2 \text{LF}_{2,2,-1}[m_1, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{48} g_1^2 g_2^2 \text{LF}_{3,1,-1}[m_1, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{96} g_1^2 g_2^2 \text{LF}_{4,1,-2}[m_1, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{96} g_1^2 g_2^2 \text{LF}_{2,1,0}[m_1, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{96} g_1^2 g_2^2 \text{LF}_{2,2,-1}[m_1, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{48} g_1^2 g_2^2 \text{LF}_{3,1,-1}[m_1, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{96} g_1^2 g_2^2 \text{LF}_{4,1,-2}[m_1, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{864} g_1^2 g_2^2 \text{LF}_{2,1,0}[m_1, m_q^{i3}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{864} g_1^2 g_2^2 \text{LF}_{2,2,-1}[m_1, m_q^{i3}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{432} g_1^2 g_2^2 \text{LF}_{3,1,-1}[m_1, m_q^{i3}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{864} g_1^2 g_2^2 \text{LF}_{4,1,-2}[m_1, m_q^{i3}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{864} g_1^2 g_2^2 \text{LF}_{2,1,0}[m_1, m_q^{i4}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{864} g_1^2 g_2^2 \text{LF}_{2,2,-1}[m_1, m_q^{i4}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{432} g_1^2 g_2^2 \text{LF}_{3,1,-1}[m_1, m_q^{i4}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{864} g_1^2 g_2^2 \text{LF}_{4,1,-2}[m_1, m_q^{i4}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{96} g_2^4 \text{LF}_{2,1,0}[m_2, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{96} g_2^4 \text{LF}_{2,2,-1}[m_2, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{5}{48} g_2^4 \text{LF}_{3,1,-1}[m_2, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{32} g_2^4 \text{LF}_{4,1,-2}[m_2, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{96} g_2^4 \text{LF}_{2,1,0}[m_2, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{96} g_2^4 \text{LF}_{2,2,-1}[m_2, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{5}{48} g_2^4 \text{LF}_{3,1,-1}[m_2, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{32} g_2^4 \text{LF}_{4,1,-2}[m_2, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{96} g_2^4 \text{LF}_{2,1,0}[m_2, m_q^{i3}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{96} g_2^4 \text{LF}_{2,2,-1}[m_2, m_q^{i3}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{5}{48} g_2^4 \text{LF}_{3,1,-1}[m_2, m_q^{i3}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{32} g_2^4 \text{LF}_{4,1,-2}[m_2, m_q^{i3}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{96} g_2^4 \text{LF}_{2,1,0}[m_2, m_q^{i4}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{96} g_2^4 \text{LF}_{2,2,-1}[m_2, m_q^{i4}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{5}{48} g_2^4 \text{LF}_{3,1,-1}[m_2, m_q^{i4}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{32} g_2^4 \text{LF}_{4,1,-2}[m_2, m_q^{i4}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{18} g_2^2 g_3^2 \text{LF}_{2,1,0}[m_3, m_q^{i3}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{18} g_2^2 g_3^2 \text{LF}_{2,2,-1}[m_3, m_q^{i3}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{9} g_2^2 g_3^2 \text{LF}_{3,1,-1}[m_3, m_q^{i3}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{18} g_2^2 g_3^2 \text{LF}_{4,1,-2}[m_3, m_q^{i3}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{18} g_2^2 g_3^2 \text{LF}_{2,1,0}[m_3, m_q^{i4}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{18} g_2^2 g_3^2 \text{LF}_{2,2,-1}[m_3, m_q^{i4}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{9} g_2^2 g_3^2 \text{LF}_{3,1,-1}[m_3, m_q^{i4}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{18} g_2^2 g_3^2 \text{LF}_{4,1,-2}[m_3, m_q^{i4}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{48} g_1^2 g_2^2 \text{LF}_{2,1,0}[m_l^{i1}, m_1] \delta_{i1i2} \delta_{i3i4} + \frac{1}{96} g_1^2 g_2^2 \text{LF}_{3,1,-1}[m_l^{i1}, m_1] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{48} g_2^4 \text{LF}_{2,1,0}[m_l^{i1}, m_2] \delta_{i1i2} \delta_{i3i4} - \frac{1}{96} g_2^4 \text{LF}_{3,1,-1}[m_l^{i1}, m_2] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{48} g_1^2 g_2^2 \text{LF}_{2,1,0}[m_l^{i2}, m_1] \delta_{i1i2} \delta_{i3i4} + \frac{1}{96} g_1^2 g_2^2 \text{LF}_{3,1,-1}[m_l^{i2}, m_1] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{48} g_2^4 \text{LF}_{2,1,0}[m_l^{i2}, m_2] \delta_{i1i2} \delta_{i3i4} - \frac{1}{96} g_2^4 \text{LF}_{3,1,-1}[m_l^{i2}, m_2] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{432} g_1^2 g_2^2 \text{LF}_{2,1,0}[m_q^{i3}, m_1] \delta_{i1i2} \delta_{i3i4} + \frac{1}{864} g_1^2 g_2^2 \text{LF}_{3,1,-1}[m_q^{i3}, m_1] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{48} g_2^4 \text{LF}_{2,1,0}[m_q^{i3}, m_2] \delta_{i1i2} \delta_{i3i4} - \frac{1}{96} g_2^4 \text{LF}_{3,1,-1}[m_q^{i3}, m_2] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{9} g_2^2 g_3^2 \text{LF}_{2,1,0}[m_q^{i3}, m_3] \delta_{i1i2} \delta_{i3i4} + \frac{1}{18} g_2^2 g_3^2 \text{LF}_{3,1,-1}[m_q^{i3}, m_3] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{432} g_1^2 g_2^2 \text{LF}_{2,1,0}[m_q^{i4}, m_1] \delta_{i1i2} \delta_{i3i4} + \frac{1}{864} g_1^2 g_2^2 \text{LF}_{3,1,-1}[m_q^{i4}, m_1] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{48} g_2^4 \text{LF}_{2,1,0}[m_q^{i4}, m_2] \delta_{i1i2} \delta_{i3i4} - \frac{1}{96} g_2^4 \text{LF}_{3,1,-1}[m_q^{i4}, m_2] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{9} g_2^2 g_3^2 \text{LF}_{2,1,0}[m_q^{i4}, m_3] \delta_{i1i2} \delta_{i3i4} + \frac{1}{18} g_2^2 g_3^2 \text{LF}_{3,1,-1}[m_q^{i4}, m_3] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{8} g_2^2 \overline{y}_d^{i4p} y_d^{i3p} \text{LF}_{3,1,-1}[\tilde{\mu}, m_d^p] \delta_{i1i2} + \frac{1}{24} g_2^2 \overline{y}_d^{i4p} y_d^{i3p} \text{LF}_{4,1,-2}[\tilde{\mu}, m_d^p] \delta_{i1i2} - \\ & \frac{1}{8} g_2^2 \overline{y}_e^{i2p} y_e^{i1p} \text{LF}_{3,1,-1}[\tilde{\mu}, m_e^p] \delta_{i3i4} + \frac{1}{24} g_2^2 \overline{y}_e^{i2p} y_e^{i1p} \text{LF}_{4,1,-2}[\tilde{\mu}, m_e^p] \delta_{i3i4} - \\ & \frac{1}{8} g_2^2 \overline{y}_u^{i4p} y_u^{i3p} \text{LF}_{3,1,-1}[\tilde{\mu}, m_u^p] \delta_{i1i2} + \frac{1}{24} g_2^2 \overline{y}_u^{i4p} y_u^{i3p} \text{LF}_{4,1,-2}[\tilde{\mu}, m_u^p] \delta_{i1i2} + \\ & \frac{1}{16} g_2^4 \text{LF}_{2,1,1,-1}[m_2, m_l^{i1}, m_q^{i3}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{8} g_2^4 m_2^2 \text{LF}_{2,1,1,0}[m_2, m_l^{i1}, m_q^{i3}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{16} g_2^4 \text{LF}_{2,1,1,-1}[m_2, m_l^{i2}, m_q^{i4}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{8} g_2^4 m_2^2 \text{LF}_{2,1,1,0}[m_2, m_l^{i2}, m_q^{i4}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{8} \overline{y}_d^{i4p} y_d^{i3p} \overline{y}_e^{i2r} y_e^{i1r} \text{LF}_{2,1,1,-1}[\tilde{\mu}, m_d^p, m_e^r] - \frac{1}{4} \tilde{\mu}^2 \overline{y}_e^{i2p} y_e^{i1p} \overline{y}_u^{i4r} y_u^{i3r} \\ & \quad \text{LF}_{2,1,1,0}[\tilde{\mu}, m_e^p, m_u^r] - \frac{1}{48} g_1^2 g_2^2 \text{LF}_{1,1,1,-1}[m_1, m_2, m_l^{i1}, m_q^{i3}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{24} m_1 m_2 g_1^2 g_2^2 \text{LF}_{1,1,1,1,0}[m_1, m_2, m_l^{i1}, m_q^{i3}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{48} g_1^2 g_2^2 \text{LF}_{1,1,1,1,-1}[m_1, m_2, m_l^{i2}, m_q^{i4}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{24} m_1 m_2 g_1^2 g_2^2 \text{LF}_{1,1,1,1,0}[m_1, m_2, m_l^{i2}, m_q^{i4}] \delta_{i1i2} \delta_{i3i4} \Big) \end{aligned}$$